

NOISE-ROBUST ENVIRONMENTAL SOUND UNDERSTANDING USING AN AUDIO FOUNDATION MODEL WITH TEXT ANCHORS

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Abstract—Environmental sound understanding, integrating sound event detection (SED), environmental sound classification (ESC), and signal-to-noise ratio (SNR) estimation in noisy environments, supports emergency response and disaster monitoring. However, three challenges remain. First, background noise degrades detection and classification performance. Second, although source separation improves noise robustness, language-driven source separation models may generate target-like signals even when the target sound is absent, causing false detections and misclassifications. Third, conventional SNR estimation directly regresses SNR from noisy mixtures without an explicit reference representation, limiting its use as a confidence measure. To address these challenges, we propose an integrated framework combining the language-driven source separation model AudioSep with the audio-language foundation model CLAP. Target and noise text embeddings are used as anchors. Source separation is combined with parameter-efficient adaptation to align acoustic embeddings of noisy mixtures with target anchors, improving noise robustness. Contrastive learning is further introduced to separate embeddings generated by mismatched source separation from the target anchors, suppressing false detections and misclassifications. A continuous, sigmoid-interpolated anchor is constructed between the target and noise anchors, and SNR is estimated from embedding similarity to this anchor. Experiments show that the proposed method achieves a recall of 51.2% and a specificity of 96.4% for SED of sirens at -20 dB SNR. For ESC over seven sound categories, it achieves an accuracy of 77.3% at -20 dB, substantially outperforming the baseline CLAP model (16.2%). The proposed anchors also support an SNR estimate that reflects the continuous ordering of SNR conditions. These results demonstrate the effectiveness of the proposed framework for robust environmental sound understanding under severe noise conditions.

Index Terms—sound event detection, environmental sound classification, SNR estimation, audio-language foundation model, LoRA

1. INTRODUCTION

Acoustic sensing complements cameras and LiDAR for detecting approaching emergency vehicles at occluded intersections or at night, and for locating survivors (voices, whistles) at disaster sites where vision is degraded or obstructed. In both settings the target sound is rarely observed in isolation—sirens are masked by traffic noise, and calls for help are masked by engine, wind, or rubble noise—so a detector that merely outputs “siren present” or “whistle present” cannot be trusted blindly: false positives and false negatives both increase as the signal-to-noise ratio (SNR) drops. We therefore argue that a practical system must jointly perform (i) sound event detection (SED) or environmental sound classification (ESC) and (ii) SNR estimation, so that the SNR acts as an explicit confidence indicator for the recognition output. We refer to this joint problem as *environmental sound understanding*. In this paper, we use the term sound event detection (SED) to refer to detecting the presence or absence of a target sound event. We evaluate the proposed framework using emergency-related sound detection and classification as representative tasks.

Realizing this faces three challenges. First, background noise directly degrades SED/ESC accuracy. Second, introducing a source-separation front-end to counter noise introduces a new failure mode,

which we call *mismatched separation*: when the query sound is absent from the mixture, a language-queried separator such as AudioSep [1] still extracts a target-like signal from the residual noise, inflating false positives. Third, conventional SNR estimators regress SNR directly from the mixture without any explicit, interpretable reference, so the estimate cannot easily be related back to the recognition decision.

To address these challenges, we integrate AudioSep as a language-queried separation front-end with the audio-language foundation model CLAP [2], [3] as the back-end, using CLAP’s frozen text-encoder outputs for the target sound and for noise as fixed *text anchors* in embedding space. We introduce two independently trained low-rank adaptation (LoRA) [4] modules in the CLAP audio encoder’s attention blocks: (i) a classification LoRA that pulls clean and noisy target embeddings toward the target anchor while pushing mismatched-separation embeddings away from it, and (ii) a SNR estimation LoRA that regresses the audio embedding toward a dynamic anchor, obtained by sigmoid-weighted interpolation between the target and noise anchors, giving the SNR estimate an explicit geometric reference.

2. RELATED WORK

2.1. Noise-robust sound classification

CNN-based classifiers trained on ESC-50 [5] or UrbanSound8K [6] achieve strong clean-condition performance, and residual connections [7] enabled deeper, more stable architectures. Multi-condition training with noise and other augmentations [5] improves robustness, but such classifiers generalize poorly to unseen noise types, over-adapt at the expense of clean accuracy, and require costly per-task retraining. An alternative is to remove noise before classification: signal-processing front-ends (beamforming, Wiener filtering) and learned separators such as Conv-TasNet [8], Demucs [9], and SepFormer [10] suppress interference ahead of the classifier, and have been shown to improve acoustic event identification when combined with microphone-array or noise-biasing front-ends [11]–[13]. However, these separators target a fixed, closed set of classes, so they do not generalize to open-vocabulary targets, and their artifacts can themselves degrade a classifier optimized independently of the front-end.

2.2. Audio foundation models and adaptation

Foundation models pretrained on large corpora substantially improve generalization over CNNs trained from scratch: PANNs [14] (CNN14/ResNet38 on AudioSet [15]), AST [16] (Transformer over log-mel patches), and BEATs [17] (self-supervised masked-patch prediction) progressively raised AudioSet and ESC-50 accuracy, although all models assumed a fixed, pre-defined label set. CLAP [2], [3], built on an HTS-AT [18] audio encoder and a RoBERTa [19] text encoder, and trained with a contrastive audio-text objective, instead maps audio and text into a shared embedding space, enabling open-vocabulary zero-shot classification via cosine similarity. However, its outputs still drift from the correct semantic pole under heavy noise.

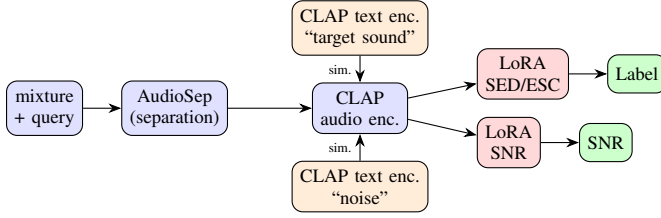


Fig. 1: Overview of the proposed architecture: AudioSep separates the target sound, CLAP audio/text encoders compute similarity to text anchors, and two independent LoRA modules perform SED/ESC and SNR estimation.

AudioSep extends this language-queried paradigm to source separation, extracting a query-matched source from an arbitrary mixture with strong zero-shot generalization, in contrast to Conv-TasNet-style separators restricted to their training domain. However, since AudioSep is trained to always produce some output, it hallucinates target-like signals from pure noise when the query is absent from the mixture—the *mismatched-separation* failure mode. To adapt frozen foundation models efficiently, parameter-efficient fine-tuning (PEFT) [20] freezes the backbone and updates only a small number of injected parameters, avoiding both the cost and forgetting risk of full fine-tuning and the limited representational adjustment of a trainable linear head alone. LoRA [4] freezes the pretrained weight matrix W and adds a parallel low-rank pair of linear layers A, B alongside it, applied to the query/key/value projections of attention layers; we build on this to adapt CLAP’s audio encoder without discarding its zero-shot capability.

2.3. SNR estimation

Classical estimators rely on signal-processing features such as spectral entropy, with auditory filter-bank compression modeling cochlear nonlinearity improving accuracy for speech in noise [21]. More recent work regresses or classifies SNR directly from deep features of the mixture [22], with regression avoiding the resolution limits of discrete bins. However, these approaches treat SNR as a physical quantity regressed in isolation, without an explicit reference tying the estimate to the target/noise semantics used by the recognizer. Our dynamic-anchor formulation targets exactly this gap: by construction, our SNR estimate is defined relative to the same shared embedding space used for classification.

3. PROPOSED METHOD

3.1. Overview

Fig. 1 shows an overview of the proposed architecture. As shown, the mixture and a natural-language query are passed to AudioSep, which extracts the target sound and thereby suppresses background noise before classification. The separated signal (as well as, in ablations, the unseparated input mixture) is encoded by the CLAP audio encoder into a 512-d embedding e . The frozen CLAP text encoder produces 512-d text anchors T_{target} and T_{noise} from natural-language descriptions of the target sound and of noise; these anchors are fixed reference points in the shared embedding space and are never updated during training. Cosine similarity between e and these anchors drives (i) SED/ESC classification and (ii) SNR estimation. Two LoRA modules, each inserted in parallel with the query/key/value projections of the audio encoder’s attention blocks, are trained for these two roles while the text encoder and the remainder of the audio encoder stay frozen; keeping the text side fixed and adapting only the audio side lets CLAP’s pretrained generalization be preserved while correcting the embedding drift that noise overlap and mismatched separation otherwise introduce. The two LoRA modules are trained independently,

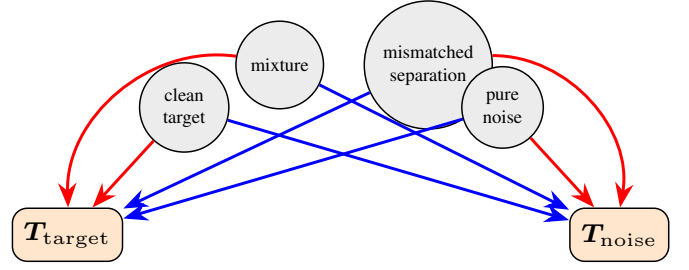


Fig. 2: Text-anchor contrastive supervision for SED (1): red arrows are pulled toward $\cos = 1$, blue arrows are pushed toward $\cos = 0$. The mismatched-separation embedding is explicitly repelled from T_{target} and attracted to T_{noise} .

one for classification and the other for SNR estimation. During inference, the two models independently encode the input audio and perform their respective tasks. Thus, the proposed framework requires two CLAP forward passes per input audio. Following standard LoRA practice, each adapter’s down-projection matrix is initialized to zero, so training begins from CLAP’s original pretrained representation and the anchor losses only gradually reshape it as training proceeds, rather than perturbing the embedding space from the outset.

3.2. Text-anchor LoRA training for SED and ESC

Figure 2 illustrates the four example types used in the SED loss. For SED with binary label $y_i \in \{0, 1\}$, the base similarity loss over a batch of N examples is

$$L = \frac{1}{N} \sum_{i=1}^N \left[(y_i - \cos(e_i, T_{\text{target}}))^2 + ((1 - y_i) - \cos(e_i, T_{\text{noise}}))^2 \right]. \quad (1)$$

The batch mixes four example types: clean/matched-separation target sounds ($y=1$), pure noise ($y=0$), noisy mixtures ($y=1$), and mismatched-separation outputs obtained by querying AudioSep with the target description on noise-only inputs ($y=0$). The total loss $L_{\text{total}} = L_{\text{target}|y=1} + L_{\text{noise}|y=0} + L_{\text{mix}|y=1} + L_{\text{mismatch}|y=0}$, where each term is an instance of (1) restricted to the corresponding subset. The mismatch term is the key novelty: it pushes mismatched-separation embeddings away from T_{target} and toward T_{noise} , directly counteracting AudioSep’s tendency to hallucinate target-like features from pure noise.

For C -class ESC with one-hot labels $y_{i,c}$, class anchors T_j , and a mismatch-prompt anchor $T_{\text{mis_prompt},i}$ (the anchor of whichever class query produced the mismatched separation for example i), we use

$$L1 = \frac{1}{N_1} \sum_{i=1}^{N_1} \sum_{j=1}^C (y_{i,c} - \cos(e_i, T_j))^2, \quad (2)$$

$$L2 = \frac{1}{N_2} \sum_{i=1}^{N_2} \cos(e_i, T_{\text{mis_prompt},i})^2, \quad (3)$$

combined as $L_{\text{total}} = L1_{\text{clean}} + L1_{\text{mix}} + L2_{\text{mismatch}}$, where $L1_{\text{clean}}$ and $L1_{\text{mix}}$ are $L1$ evaluated respectively on clean/matched-separation examples and on noisy mixtures, and $L2_{\text{mismatch}}$ is $L2$ on mismatched-separation examples. In practice, mismatch-prompt anchors $T_{\text{mis_prompt},i}$ are obtained by issuing all C class queries to AudioSep for a given recording and treating the $C-1$ queries that do not match the true class as mismatched-separation sources; this mirrors the unknown-target inference setting evaluated in Section 4 (condition B), where the true class is unknown beforehand and every class description is queried against the same mixture. In both the

SED and ESC formulations, the classification LoRA is trained jointly on clean, mixed, and mismatched-separation examples within each batch, so that a single adapter learns to be simultaneously robust to noise overlap and to AudioSep’s hallucinated separations, rather than addressing the two failure modes with separate models.

3.3. SNR estimation via dynamic anchor interpolation

To give SNR estimation an explicit reference, we map a physical SNR value to an interpolation ratio $\lambda \in (0, 1)$ via a sigmoid,

$$\lambda = \sigma(0.23 \cdot \text{SNR}) = \frac{1}{1 + e^{-0.23 \cdot \text{SNR}}}, \quad (4)$$

with the slope chosen so that the operating range of -20 to $+20$ dB spans the sigmoid’s sensitive region. We then build a dynamic text anchor by interpolating between the target and noise anchors and re-normalizing,

$$\mathbf{T}_{\text{dynamic}} = \text{Normalize}(\lambda \cdot \mathbf{T}_{\text{target}} + (1 - \lambda) \cdot \mathbf{T}_{\text{noise}}). \quad (5)$$

As SNR increases, $\mathbf{T}_{\text{dynamic}}$ moves continuously from the noise anchor toward the target anchor, forming a one-dimensional geometric structure in the shared embedding space. Concretely, at $\text{SNR} = +10$ dB, $\lambda \rightarrow 1$, so $\mathbf{T}_{\text{dynamic}}$ nearly coincides with $\mathbf{T}_{\text{target}}$; at $\text{SNR} = 0$ dB, $\lambda = 0.5$ places $\mathbf{T}_{\text{dynamic}}$ exactly midway between the two poles; and at $\text{SNR} = -10$ dB, $\lambda \rightarrow 0$ and $\mathbf{T}_{\text{dynamic}}$ nearly coincides with $\mathbf{T}_{\text{noise}}$. A dedicated SNR LoRA is trained with

$$L_{\text{snr}} = \frac{1}{N} \sum_{i=1}^N (1 - \cos(\mathbf{e}_i, \mathbf{T}_{\text{dynamic},i}))^2, \quad (6)$$

so that the mixture embedding is pulled toward the dynamic anchor corresponding to its true SNR. At inference, SNR is estimated by evaluating cosine similarity against a discrete set of anchors $S = \{-20, -10, -5, 0, 5, 10, 20\}$ dB and selecting the closest one.

4. EXPERIMENTS

4.1. SED: emergency-vehicle siren detection

We use sireNNet [23] (1,675 3-s clips of ambulance/fire-truck/police sirens and traffic noise; 839 train / 209 test clips) and evaluate recall on siren-containing clips (clean and mixed with traffic noise at $\text{SNR} \in \{0, -5, -10, -20\}$ dB) and specificity on traffic-noise-only clips. Denoting the siren-containing and traffic-noise-only test subsets by $\mathcal{D}_s$ and $\mathcal{D}_n$ and the detector output by $\hat{y}(x) \in \{0, 1\}$,

$$\text{Recall} = \frac{|\{x \in \mathcal{D}_s : \hat{y}(x)=1\}|}{|\mathcal{D}_s|}, \quad \text{Spec.} = \frac{|\{x \in \mathcal{D}_n : \hat{y}(x)=0\}|}{|\mathcal{D}_n|}, \quad (7)$$

with recall computed separately within each SNR condition and specificity computed once over the pooled traffic-noise clips. Table 1 compares all eight configurations: frozen CLAP zero-shot, CLAP with LoRA trained on clean sirens/noise only and then also with mixture training (L_{mix}), AudioSep+CLAP without any retraining (Vanilla), and AudioSep+CLAP with LoRA progressively adding multi-condition training (L_{mix}) and mismatch suppression (L_{mismatch}), culminating in the proposed model that combines both losses.

4.2. ESC: 7-class disaster-relevant sound classification

We select 7 ESC-50 [5] classes relevant to disaster scenarios (car horn, chainsaw, crow, dog, engine, fire, siren; 1,123 train / 167 test 1-s segments, each 5-s ESC-50 clip split into 1-s windows) and mix them with drone noise recorded by a microphone array embedded in an unmanned aerial vehicle, taken from DREGON [24]. Per class, the split is close to balanced, so the low-SNR accuracy drops reported below are not an artifact of class imbalance. We evaluate under (A)

Table 1: SED results: recall [%] per input SNR condition and specificity [%] on traffic noise (No. follows the numbering used in the text.)

No.	Model	AudioSep	SNR0	SNR-5	SNR-10	SNR-20	Spec.
1	CLAP zero-shot [2]	-	93.6	85.6	68.8	7.2	98.8
2	+LoRA [4]	-	97.6	92.8	76.8	10.4	100
3	+LoRA (L_{mix})	-	99.2	99.2	93.6	35.2	95.2
4	CLAP Vanilla	✓	95.2	96.8	92.8	46.4	92.9
5	+LoRA	✓	99.2	98.4	95.2	60.8	91.7
6	+LoRA (L_{mix})	✓	99.2	99.2	99.2	71.2	86.9
7	+LoRA (L_{mismatch})	✓	96.8	94.4	84.8	29.6	98.8
8	Proposed ($L_{\text{mix}}+L_{\text{mismatch}}$)	✓	99.2	99.2	99.2	51.2	96.4

Table 2: ESC condition A (matched separation query) accuracy [%]. (MC means multi-condition training.)

No.	Model	Clean	SNR0	SNR-5	SNR-10	SNR-20
1	ResNet [7] (MC)	74.3	61.1	56.3	47.9	41.9
2	AudioSep+ResNet (MC)	71.3	71.3	70.1	70.1	72.5
3	CLAP zero-shot [2]	82.0	49.7	38.3	27.5	16.2
4	CLAP Extended	71.3	16.2	14.4	13.2	13.8
5	CLAP Extended (MC)	79.0	65.9	57.5	52.1	43.1
6	CLAP LoRA [4]	92.2	59.9	42.5	30.5	17.4
7	CLAP LoRA (L_{mix})	91.6	91.6	87.4	80.8	65.9
8	AudioSep+CLAP Vanilla	86.8	66.5	58.1	45.5	22.2
9	AudioSep+CLAP LoRA	94.0	82.0	77.3	64.7	32.3
10	+ (L_{mismatch})	94.6	76.7	67.1	56.9	20.4
11	+ (L_{mix})	92.8	94.0	88.6	85.6	64.7
12	Proposed (both losses)	91.0	92.2	91.0	86.8	77.3

a matched condition, where AudioSep is queried with the ground-truth class so that separation always targets the correct source, and (B) a more realistic unknown-target condition, where the true class is not known in advance and all 7 class descriptions are issued to AudioSep as separate queries, so all 7 resulting separated channels (6 of which are necessarily mismatched, since only one class is actually present) are passed to the classifier. For each of the seven separated signals, cosine similarity is computed with all seven class text anchors, resulting in 49 similarity scores in total. The class corresponding to the highest similarity score is selected as the final predicted class. Table 2 reports accuracy for condition A, contrasting CLAP LoRA tuning against a from-scratch ResNet classifier, an extended-learning (frozen backbone + trainable linear head) baseline, and their AudioSep-fronted variants; Table 3 reports condition B, the more realistic setting where mismatched separation is unavoidable at test time.

5. RESULTS AND DISCUSSION

5.1. SED

Results from Table 1 show that CLAP zero-shot (No. 1) collapses to 7.2% recall at -20 dB, confirming severe noise vulnerability. Even without the AudioSep front-end, LoRA adaptation alone helps: adding L_{mix} to a separation-free LoRA (No. 3) raises recall at -10 dB from 76.8% (No. 2, clean/noise-only training) to 93.6%, but recall at -20 dB still plateaus at 35.2%, far below the 71.2% reached once AudioSep is introduced (No. 6). This indicates that adaptation and front-end separation address complementary aspects of the noise-robustness problem, with separation being necessary to reach usable recall in the most severe conditions. Adding AudioSep and L_{mix} raises low-SNR recall substantially but lowers specificity from 98.8% to 86.9%, i.e., mismatched separation induces false positives. Adding L_{mismatch} alone (No. 7) restores specificity to 98.8% but caps recall improvements. The proposed combination retains 96.4% specificity while achieving 51.2% recall at -20 dB. Comparing the Vanilla AudioSep pipeline (No. 4, specificity 92.9%) with the same pipeline after adding L_{mix} (No. 6, specificity 86.9%) isolates where this loss originates: pulling noisy-mixture embeddings toward the target anchor during training also nudges purely noisy, target-absent embeddings closer to that same anchor at test time, i.e., multi-condition training

Table 3: ESC condition B (7-way separation, unknown target) accuracy [%].

No.	Model	Clean	SNR0	SNR-5	SNR-10	SNR-20
1	CLAP zero-shot	82.0	49.7	38.3	27.5	16.2
2	AudioSep+CLAP (Vanilla)	77.8	61.7	55.1	40.1	17.4
3	AudioSep+CLAP Extended (MC)	87.4	82.0	82.6	74.3	47.9
4	AudioSep+CLAP LoRA	88.6	58.1	53.3	40.7	19.2
5	+ (L_{mismatch})	96.4	64.1	49.7	41.3	21.6
6	+ (L_{mix})	82.0	67.1	67.1	64.1	49.1
7	Proposed (both losses)	92.8	88.0	88.0	80.2	62.9

by itself trades specificity for recall. This is exactly the side effect that L_{mismatch} is introduced to counteract, and No. 8 shows that the two losses combined avoid this trade-off rather than merely balancing it.

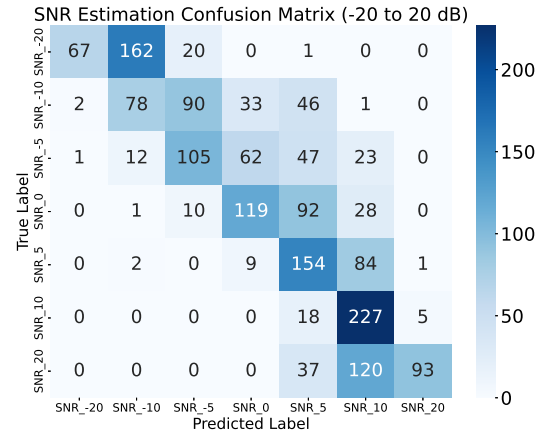
5.2. ESC

In condition A (Table 2), CLAP zero-shot (No. 3) outperforms ResNet trained from scratch (No. 1), confirming the value of large-scale pretraining, but both collapse under noisy conditions. Placing AudioSep (No. 2) ahead of the same ResNet classifier and retraining it with multi-condition data yields a consistent 70.1–72.5% accuracy over the range of input SNRs. This shows that a fixed-capacity CNN behind a separation front-end can become largely insensitive to input SNR, but it never exceeds roughly 73% even in clean or matched conditions. No. 4 Extended learning (frozen backbone, trainable linear head) cannot adapt CLAP’s internal representation and even falls below the untuned model on clean audio (71.3% vs. 82.0%), whereas LoRA raises clean accuracy to 92.2%, confirming that adapting the attention layers directly, rather than only the output head, is necessary to exceed the representational ceiling of a fixed backbone. Adding AudioSep (No. 9) roughly doubles SNR−20 accuracy at matched LoRA settings (17.4%→32.3%), and combining $L_{1\text{mix}}$ with $L_{2\text{mismatch}}$ (No. 12) yields the best low-SNR accuracy (77.3% at −20 dB) among all condition-A configurations.

In the more realistic condition B (Table 3), where 6 of 7 separated channels are mismatched at inference, the ordering changes and the impact of mismatched separation on clean accuracy becomes directly visible: the untuned Vanilla pipeline (condition A - No. 8) drops from 86.8% clean accuracy in condition A’s matched setting (condition B - No. 2) to 77.8% here, purely because six spurious separated channels are now fed to the classifier alongside the correct one. L_{mismatch} alone (No. 5) now yields the best clean accuracy (96.4%) because it directly targets the mismatched channels that dominate the input distribution at test time, but its low-SNR gains remain modest (21.6% at −20 dB) since it does not teach robustness to real noise overlap. L_{mix} alone (No. 6) substantially improves mid-to-low-SNR robustness instead, reaching 49.1% at −20 dB at the cost of clean accuracy falling back to 82.0%. Combining both losses is synergistic rather than additive: the proposed model (No. 7) reaches 62.9% at −20 dB while also recovering clean accuracy to 92.8%, exceeding either loss used alone at every SNR level, indicating that the noise-robust representation learned via L_{mix} is a prerequisite for L_{mismatch} to correctly separate true mismatches from genuine detections once both losses act on a shared, jointly-trained embedding space. Across both SED and ESC, the same pattern holds: mismatch suppression and mixture-based multi-condition training address distinct failure modes and are most effective when trained together rather than in isolation.

5.3. SNR estimation

We evaluate the dynamic-anchor SNR estimator described in Section 3 alongside the SED task, by computing, for each of the seven true-SNR conditions $S = \{-20, -10, -5, 0, 5, 10, 20\}$ dB, the confusion

**Fig. 3:** SED-task SNR estimation confusion matrix: true SNR (rows) vs. predicted SNR bin (columns), out of 250 test examples per row.

between the true SNR bin and the bin whose anchor is closest to the mixture embedding in cosine similarity. Figure 3 shows the resulting 7×7 confusion matrix.

True and predicted SNR are strongly correlated along the diagonal of Fig. 3. It indicates that the sigmoid-interpolated anchors preserve the ordering of the underlying continuous SNR scale. Exact-match counts peak at 5 dB (154/250) and 10 dB (227/250), where the sigmoid’s slope, and hence the angular separation between neighboring anchors, is steepest, and are lower at the extremes −20 dB (67/250) and +20 dB (93/250), where the sigmoid saturates and adjacent anchors are compressed together in the embedding geometry. The off-diagonal mass is not uniform across conditions: at the extremes of the operating range, misclassifications are concentrated in the immediately neighboring bin (162 of 183 at −20 dB; 120 of 157 at +20 dB), whereas near the middle of the range misclassifications are more dispersed across multiple neighboring bins.

6. CONCLUSION

We introduced environmental sound understanding as the joint problem of noise-robust SED/ESC and interpretable SNR estimation, and proposed a text-anchor-guided LoRA adaptation of CLAP combined with an AudioSep front-end. Two independent LoRA modules address noise overlap and AudioSep’s mismatched separation for classification, while a sigmoid-interpolated dynamic anchor supports SNR estimation with an explicit geometric reference. On siren SED and 7-class disaster-relevant ESC, the proposed method substantially improves low-SNR performance while preserving specificity/clean accuracy, and the anchor-based SNR estimate tracks the true SNR ordering even where it cannot resolve the exact dB value. These results were obtained on synthetically mixed, single-target recordings. Future work includes expanding the range of target sound classes to evaluate the generalization of the proposed method to a broader set of environmental sounds, and extending the evaluation to a wider variety of noise types to assess robustness against changes in background noise conditions. We also plan to compare the proposed approach with other audio foundation models to further investigate the contribution of text anchoring. In addition, future work includes extending the framework to mixtures of three or more overlapping sound sources and validating the full pipeline in field trials with real emergency-vehicle and disaster-response audio.

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